AI-Capstone Advanced Proposal
Group 13

Task: Post-Meal Tabletop Cleanup with Tableware Clearing and Surface Wiping

1. Task Objectives

   This advanced-level project aims to study a post-meal tabletop cleanup task using a single robot arm. The robot is required to clear target tableware from a dining table, place the tableware into a tray, retrieve a wiping cloth, and wipe a specified dirty region on the table. At the same time, the robot must avoid disturbing non-target objects, such as a tissue box or a vase, that may appear in the workspace but should not be collected or moved.

   Compared with the Entry-level cutlery arrangement task, this task extends the problem from simple object placement to a more complete household cleanup scenario. The robot must distinguish between different object roles, complete multiple dependent subtasks in the correct order, use a tool for surface cleaning, and satisfy safety constraints. Therefore, the task is formulated as a constrained multi-stage manipulation problem involving tableware clearing, tool use, surface coverage, and disturbance avoidance.

   The main task objectives are:

   1. Move target tableware objects into a designated tray.
   2. Retrieve and use a cloth to wipe a dirty region on the table.
   3. Avoid disturbing protected non-target objects.
   4. Avoid dropping objects or pushing objects off the table.
   5. Complete the task within a fixed episode horizon.
   6. Motivation and Problem Formulation

2. Motivation and Problem Formulation

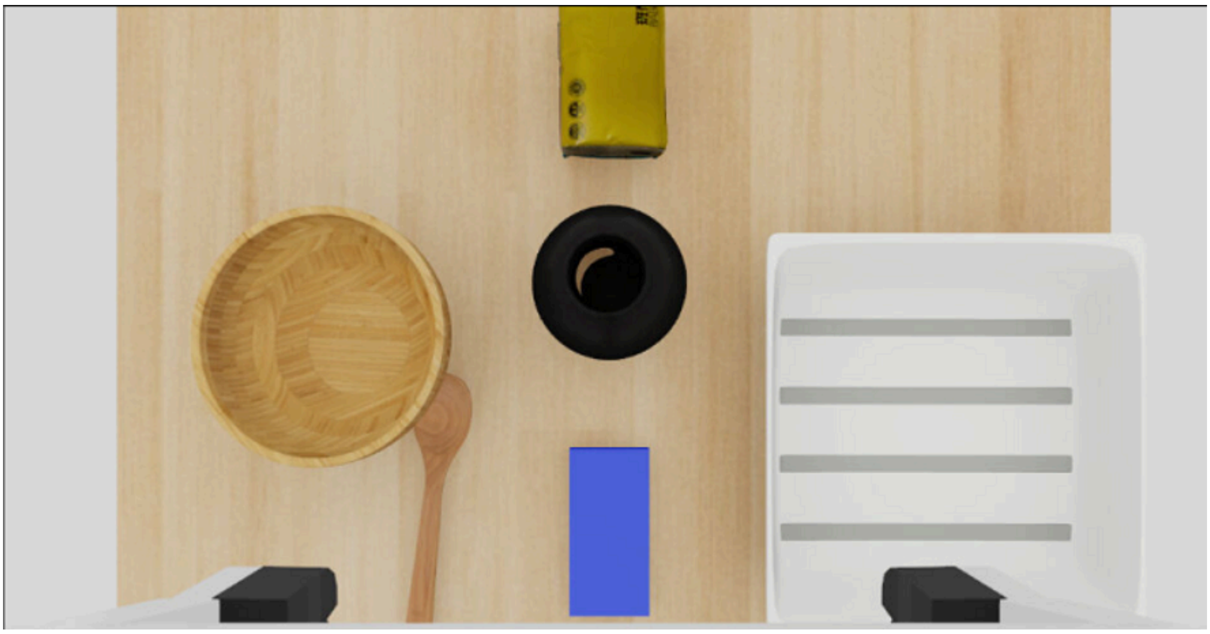
   Dining cleanup is a common and repetitive household activity. A robot capable of clearing tableware and wiping a table surface could be useful in home assistance, elder care, mobility assistance, and service robotics. While the Entry-level project focuses on arranging utensils before a meal, this advanced task focuses on cleaning up the table after a meal.

   The proposed task is designed to evaluate whether a robot policy can perform role-aware and sequential manipulation. The scene contains several object types with different roles. Bowl and spoon are target tableware objects that should be collected. The tray is the receptacle for cleared tableware. The cloth is a tool used for wiping the table. Tissue box and vase are protected objects that should remain in place.

A successful robot policy must understand that not all objects in the scene should be manipulated in the same way. Moving a bowl or spoon into the tray is correct, but moving a tissue box or vase is considered a failure. The robot must also perform the task in a meaningful order: it should clear the tableware before wiping the dirty region, because uncleared objects may block the wiping path or be pushed during cleaning.

3. Datasets

The dataset will be generated in simulation. We will not use the provided USD files from the Entry-level tasks. All objects and scene assets used in this advanced task will be independently created or downloaded from external sources with appropriate licenses. The USD files we will use in the advance level is stored at: AI_Capstone_objects and we have put them into the scene:



The dataset will contain episodes of the robot completing the cleanup task. Each episode will include robot observations, robot actions, object states, and the final task outcome. The demonstrations may be generated using expert scripted demonstrations and, if necessary, human teleoperation demonstrations. Expert demonstrations provide consistent successful trajectories, while teleoperation demonstrations can help capture correction behaviors and more realistic interactions.

The dataset will include variations in the tabletop layout. Target tableware objects will be initialized at different positions and orientations on the table. The dirty region may also be placed at different locations within the reachable workspace. Protected objects will be placed near the workspace to evaluate whether the robot can avoid disturbing them during clearing and wiping.

The initial scope will focus on a simplified scene with two target tableware objects, one tray, one cloth, and one or two protected objects. If this version is successful, the scene can be extended by adding more tableware objects or increasing the layout diversity.

4. Simulation Environment

The simulation environment will contain a dining table, a single robot arm, target tableware, a tray, a wiping cloth, protected objects, and a specified dirty region. The robot must interact with the target objects and cloth while avoiding unnecessary contact with protected objects.

The environment will include the following object roles:

Object Type	Role in the Task
Bowl and spoon	Target tableware objects to be cleared
Tray	Receptacle for cleared tableware
Cloth	Tool used for wiping the table
Tissue box and vase	Protected non-target objects

The dirty region will be represented as a predefined region on the tabletop. Instead of physically simulating liquid, dust, or food stains, we will use an abstract surface-coverage criterion to determine whether the wiping behavior covers enough of the target region. This design makes the task easier to reproduce and evaluate consistently.

The environment will be designed as a new advanced-level task and will not directly reuse the Entry-level environment or provided Entry-level USD assets. Similar object categories may be used, but the assets will be independently prepared.

5. Methods

The proposed method follows a staged manipulation strategy. The robot first clears the target tableware objects and then performs table wiping. This decomposition reflects the physical structure of the task and reduces ambiguity during policy learning and evaluation.

In the tableware clearing stage, the robot must identify the target tableware objects, grasp them, move them into the tray, and release them safely. This stage evaluates object-specific grasping, transport stability, and constrained placement.

In the table wiping stage, the robot retrieves the cloth and wipes the dirty region. The wiping motion should cover a sufficient portion of the dirty area while avoiding protected objects. This stage evaluates tool use, surface coverage, and collision-aware motion in a cluttered tabletop workspace.

We plan to train an imitation-learning policy using the generated demonstrations. The main baseline will use the policy training pipeline provided in the course framework. If time permits, we may compare different policy settings or dataset variants, such as training with only scripted demonstrations versus training with additional teleoperation demonstrations.

To keep the implementation feasible, the first version will use a simplified cloth or wiping tool representation. More realistic cloth dynamics may be considered only as a future extension.

6. Expected Outcomes

The expected outcome is a reproducible advanced-level simulation task for post-meal tabletop cleanup. We expect the robot to complete the simplified cleanup task by collecting target tableware into a tray and wiping the designated dirty region while preserving protected objects.

The expected deliverables include:

1. A custom simulation environment for tabletop cleanup.
2. Independently prepared scene and object assets.
3. A generated demonstration dataset.
4. A trained policy checkpoint.
5. Quantitative evaluation results.
6. Qualitative rollout screenshots or videos.
7. Failure analysis for different task stages.

We expect the policy to perform better in simpler layouts where target objects and protected objects are clearly separated. More cluttered layouts are expected to reveal challenges in grasping, sequencing, wiping coverage, and disturbance avoidance.

7. Evaluation Criteria

The task will be evaluated using four main criteria: tableware placement, wiping coverage, protected-object stability, and object safety.

Tableware placement success measures whether all target tableware objects are placed inside the tray. A rollout fails this criterion if any target tableware object remains outside the tray or falls off the table.

Wiping coverage measures whether the cloth covers a sufficient portion of the predefined dirty region. The wiping stage is successful if the covered area exceeds a predefined threshold.

Protected-object stability measures whether non-target objects remain close to their initial poses. A rollout fails this criterion if the tissue box, vase, or other protected object is moved beyond an allowed tolerance.

Object safety measures whether any object falls off the table or undergoes severe unintended disturbance. This prevents the robot from achieving placement or wiping by using unsafe or unrealistic motions.

An episode is considered successful only if all of the following conditions are satisfied:

1. All target tableware objects are placed in the tray.
2. The dirty region is wiped above the required coverage threshold (ex: 80%).
3. All protected objects remain within the allowed displacement tolerance.
4. No task object falls off the table or undergoes severe unintended disturbance.
5. The task is completed within the maximum episode length.

We will report the overall success rate over a fixed number of rollout trials. We will also report subtask-level results, such as tableware clearing success, wiping success, and protected-object stability, to better understand the failure modes.

8. Anticipated Challenges and Possible Solutions

The first challenge is multi-stage task sequencing. The robot must clear the tableware before wiping the table. If wiping starts too early, the cloth may collide with the tableware or push objects into protected regions. To address this issue, we will use a staged task design and evaluate the clearing and wiping subtasks separately.

The second challenge is object-specific grasping. A bowl and a spoon have different shapes and require different grasping behaviors. The spoon is thin and sensitive to gripper alignment, while the bowl may require stable contact around its rim or side. To reduce difficulty in the first version, we will use a limited number of object types and avoid highly cluttered initial layouts.

The third challenge is role distinction. The robot must distinguish target tableware, tools, receptacles, and protected objects. The correct behavior depends not only on object geometry but also on task role. We will define object roles clearly in the environment and evaluate whether the robot avoids manipulating protected objects.

The fourth challenge is wiping evaluation. Wiping is an area-coverage task rather than a point-placement task. To make evaluation reproducible, we will use a predefined dirty region and a coverage-based success criterion instead of simulating complex liquid or stain physics.

The fifth challenge is disturbance avoidance. Protected objects may be close to the robot's workspace, so the robot must avoid collisions while carrying objects and wiping the table. We will begin with a simple layout and gradually increase the difficulty by placing protected objects closer to the workspace.

The final challenge is demonstration quality. Since the task contains multiple stages, poor or inconsistent demonstrations may cause ambiguous training data. We will inspect generated trajectories and separate failure cases by subtask to support error analysis.

9. Current Scope and Future Extensions

The current scope focuses on a single-arm dining cleanup scenario with two target tableware objects, one tray, one cloth, and one or two protected tabletop objects. This scope is sufficient to study sequencing, role-aware manipulation, constrained placement, tool use, and wiping coverage.

Future extensions may include adding more tableware objects, randomizing the positions of the tray and protected objects, increasing scene diversity, or using more realistic deformable cloth dynamics. Another possible extension is to design adaptive wiping policies that plan coverage motions based on the current table layout instead of following a fixed wiping pattern.

If the full cleanup task proves too difficult, we will use a reduced version as a fallback: the robot first clears target tableware into the tray and then performs a simplified wiping motion over a fixed dirty region. This fallback still preserves the main research focus of multi-stage table cleanup while keeping the task feasible within the project timeline.